

Emily Xie

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Education

Carnegie Mellon University - Bachelor of Science in Computer Science and Information Systems Expected May 2025
GPA: 4.0/4.0 | Selected Coursework: Distributed Systems, Machine Learning, Computer Graphics, Application Design and Development, Database Design and Development, Data Structures and Algorithms, Creative Coding

Experience

Software Engineer Intern - Apple May 2024 - August 2024

- Integrated new feature in audio stack to unify code architecture across Apple's ecosystem.
- Completed end-to-end feature integration process including design, implementation, reviews, testing, and merging.
- Migrated functionality between daemons; gained experience in security concepts, systems, and C.
- Coordinated cross-functional meetings to align on code flows before implementing changes across 4 repositories.

Software Engineer Intern - NASA Jet Propulsion Laboratory June 2023 - August 2023

- Developed planning and execution software for Europa Clipper – NASA mission exploring Jupiter's icy moon, Europa.
- Finished backend programming tasks in Python, GraphQL, and Java to simulate Europa Clipper spacecraft and improve operations workflow; emitted 10,000+ lines of code and presented 4 technical reports.

Visiting Scholar/Research Intern - UCSF Department of Radiology June 2020 - August 2023

- Developed machine learning pipeline in Python and Tensorflow to analyze brain imaging data (Prof. Yan Li).
- Created statistical analysis pipeline for osteoarthritis MRI images in MATLAB (Prof. Valentina Pedoia).
- Publications:

“Evaluation of Deep Learning Models for Quality Control of MR Spectra”, S. Vaziri, H. Liu, **E. Xie**, et al., *Frontiers In Neuroscience*, 2023, doi: 10.3389/fnins.2023.1219343.

“Large-Scale Analysis of Meniscus Morphology as Risk Factor for Knee Osteoarthritis”, K. Gao, **E. Xie**, et al., *Arthritis & Rheumatology*, 2023, doi: 10.1002/art.42623.

Research Assistant - CMU Augmented Perception Lab September 2022 - June 2023

- Conducted research on sound spatialization in AR/VR environments and UI layout optimization.
- Created 3 experimental setups in Unity, facilitated over 30 studies with participants, and examined resulting data to design 2 optimization algorithms in Python.

Software Developer Intern - Sandia National Laboratories May 2022 - August 2022

- Researched data generation/subsampling methods in Java to optimize runtime of gas attack simulation software.
- Reduced input data to 5% of original size while maintaining equivalent results; showcased results at 2 symposiums.

Projects/Activities

Teaching Assistant - CMU, 10-601 Intro to Machine Learning August 2023 - May 2024

- Led homeworks on neural networks, wrote exam problems, and mentored students at office hours.

Innovation Team Lead - TEDxCMU August 2021 - Present

- Experimented with Typescript, Next.js, and Three.js to create interactive web experiences and physical installations.

Solo Game Developer - Jazzcats! August 2021

- Architected, drew, and published full-stack iOS game about jazz and cats.

Skills

Python, Javascript, Typescript, React, HTML/CSS, Pytorch, Docker, Java, Go, C/C#/C++, Objective-C, Swift, Ruby, SQL, MongoDB, GraphQL, Firebase, Unity, Blender, Maya, Three.js, p5.js